Answer following question

1 :- When do we go for Chi square test ?

Ans :- Chi-Square Test When do we use Chi-Square Test?

The Chi-square test is used when we want to analyze categorical (non-numerical) data and check whether there is a significant relationship or difference between variables.

Situations to Use Chi-Square Test Testing Independence of Variables: To check whether two categorical variables are related or independent. Example: Relationship between gender and product preference. Goodness of Fit Test: To check whether observed data fits an expected distribution. Example: Whether dice outcomes follow equal probability. Comparing Proportions: To compare frequencies or counts in different categories. Example: Number of students choosing different streams.

Conditions for Applying Chi-Square Test Data should be in frequency (count) form Categories should be mutually exclusive Sample size should be large enough (expected frequency ≥ 5 preferred) Observations should be independent

Example

Suppose we want to check whether smoking habit and health condition are related:

H_0: No relationship between smoking and health H_1: There is a relationship

Conclusion

We use the Chi-square test when dealing with categorical data to test independence, association, or goodness of fit between variables.

2 :- Do we use Chi square test for independence ? Explain How ?

Ans :- Chi-Square Test for Independence Do we use Chi-square test for independence?

Yes, the Chi-square test is used to check whether two categorical variables are independent or associated.

Explanation

The test compares the observed frequencies (O) with the expected frequencies (E) under the assumption that the variables are independent.

Steps Involved Form Hypotheses: H_0: The two variables are independent H_1: The two variables are dependent (associated) Create Contingency Table: Arrange data in rows and columns (e.g., Gender vs Preference) Calculate Expected Frequencies:

E=(Row Total×Column Total)/Grand Total

Compute Chi-Square Statistic: χ2=∑E(O−E)^2/E Decision Rule: Compare calculated χ^2 value with table value OR Use p-value approach If p-value < α ⟶ Reject H_0

Example

To check whether gender and product choice are independent:

Collect data ⟶ form contingency table Apply Chi-square formula If H_0 is rejected ⟶ variables are dependent

Conclusion

The Chi-square test for independence helps determine whether two categorical variables are related or independent by comparing observed and expected frequencies.

3 :- Explain role of Critical value and p-value in Chi square test.

Ans :- Role of Critical Value and p-value in Chi-Square Test

In a Chi-square test, both critical value and p-value are used to decide whether to accept or reject the null hypothesis (H_0).

1. Role of Critical Value

Definition: The critical value is the threshold value obtained from the Chi-square table based on:

Level of significance (α) Degrees of freedom (df)

Role in Decision Making:

If calculated χ^2 > critical value ⟶ Reject H_0 If calculated χ^2 ≤ critical value ⟶ Accept H_0

2. Role of p-value

Definition: The p-value is the probability of obtaining the observed result assuming H_0 is true.

Role in Decision Making:

If p-value < α ⟶ Reject H_0 If p-value ≥ α ⟶ Accept H_0

Conclusion

Both critical value and p-value help in hypothesis testing:

Critical value gives a cut-off point p-value gives the exact significance level

4 :- Can we use chi square test to show existence of correlation between 2 features in a dataset ? How will you interpret test result to indicate correlation presence.

Ans :- Using Chi-Square Test to Show Correlation Between Two Features 1 :-) Can we use it?

Yes, the Chi-square test of independence can be used to check whether there is a relationship (association/correlation) between two categorical features in a dataset. 2) How to Use It

Step 1: Form Hypotheses

H_0: The two features are independent (no correlation) H_1: The two features are dependent (correlated/associated)

Step 2: Create Contingency Table

Arrange the frequency of observations for each combination of feature categories.

Step 3: Compute Chi-Square Statistic

$\chi 2=\sum E(O-E)^2/E$ O = observed frequency E = expected frequency (assuming independence) Step 4: Compare with Critical Value or p-value

If χ^2 > critical value OR p-value < α ⟶ Reject H_0 ⟶ Features are associated If χ^2 ≤ critical value OR p-value ≥ α ⟶ Accept H_0 ⟶ Features are independent 3) Interpretation Reject H_0 ⟶ There is evidence of correlation (association) between the two features. Accept H_0 ⟶ No evidence of correlation; the features are independent.

Example: Check if Gender and Product Choice are related:

H_0: Gender and Product Choice are independent H_1: Gender and Product Choice are associated

If test result:

p-value = 0.02 and α = 0.05 ⟶ Reject H_0 ⟶ Gender influences product choice

Conclusion

Chi-square test can indicate the presence of correlation between categorical features by testing whether the variables are independent or associated.

5 :- What is degrees of freedom and how to calculate it while performing chi square test.

Ans :- Degrees of Freedom in Chi-Square Test

1. Definition

Degrees of freedom (df) in a Chi-square test refers to the number of independent values or categories that can vary while performing the test.

It is used to determine the critical value from the Chi-square table. Higher df → more categories → wider distribution.

2. How to Calculate

a) For Chi-square Test of Independence (Contingency Table): df=(Number of rows−1)×(Number of columns−1) Rows: Number of categories of first variable Columns: Number of categories of second variable Example: A table of Gender (2 categories) vs Product Choice (3 categories) df=(2−1)×(3−1)=1×2=2

b) For Chi-square Goodness of Fit Test: df=(Number of categories−1−number of parameters estimated) Usually, no parameters estimated → df = categories − 1

3. Role in Chi-Square Test Determines the shape of the Chi-square distribution. Used to find critical value for decision-making. Ensures the test accounts for independent information only.

Conclusion

Degrees of freedom (df) quantify the number of independent comparisons in the data and are essential for interpreting the Chi-square statistic correctly.